SketchRAG: Approximate Streaming Summaries as Retrieval Context for Agentic AI

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Abstract

Agents that read continuous streams cannot afford to keep and search every chunk forever. **SketchRAG** maintains a bounded, query-independent summary of the stream: semantic cells with extractive digests, per-cell Count-Min Sketches, Misra-Gries heavy hitters and streaming IDF, all updated in a single pass. It uses this summary as the retrieval layer. The Count-Min error bound, checked against each digest, gates when the agent must read raw evidence from cold storage. On 350 LongBench / ∞-Bench questions answered by `qwen2.5:7b-instruct` through Ollama, the gated configuration scores **41.7**. That beats full context (38.1), dense RAG (35.8), SARA-lite (38.2) and AMBER-lite (32.0), while using the fewest LLM context tokens of any system (1,213 per question, 8.5% of full context). BM25 RAG remains the strongest system (46.0). The gate is well calibrated: on the 54% of questions it escalates, raw evidence lifts the score from 30.0 (digest only) to 51.5. On the questions it trusts, always reading raw evidence would add only 0.8 points. On ∞-Bench KV retrieval the gate turns 6.7 (digest only) into 96.7. With Phi-3.5-mini as the shared model, SketchRAG matches the official EM-LLM reproduction on the six LongBench tasks EM-LLM could complete on Kaggle T4s. On an 8.5M-word stream, the searchable memory stays at 1–4 MB against a 146 MB dense index. At that scale, recall drops (0.29 vs 0.44), which is the main limitation we report.

1. Introduction

LLM agents increasingly read information that arrives continuously: logs, transcripts, documents appended over time. Conventional retrieval-augmented generation (RAG) keeps every chunk and a vector for each one, so its searchable index grows linearly with the stream. Long-context and episodic-memory methods (EM-LLM) avoid re-reading everything but still keep per-event key/value state for the whole history.

SketchRAG asks whether a *bounded*, continuously maintained approximate summary of the stream can serve as the retrieval layer itself. It combines classic streaming-data synopses (Count-Min Sketch, Misra-Gries heavy hitters, reservoir sampling, streaming clustering) with an agentic fallback, so the agent answers from the summary when that is safe and reads raw evidence only when the sketch says it cannot vouch for the answer.

Central research question. Can approximate streaming summaries provide a memory-efficient retrieval context for agentic AI while keeping answer quality comparable to full-context and RAG approaches?

Contributions

1. **Summaries maintained before any query.** Compression is applied to the *stream* as it arrives, not to retrieved passages after a query (SARA) and not at query time (AMBER). The hot memory has a fixed size whatever the stream length.
2. **An error-bound-gated fallback.** The Count-Min Sketch's guarantee ($f(x) \le est(x) \le f(x) + \varepsilon N$ with probability $1-\delta$) and the digest contents decide, per retrieved cell, whether the summary can be trusted. Cells that fail are escalated to raw evidence in cold storage. This turns a classic sketch guarantee into a calibrated "is the summary enough?" signal for the agent (RQ4).

2. Method

Architecture

The document or stream is split into chunks of about 150 word-units, where long machine tokens such as UUIDs are charged per 6 characters. Each chunk is embedded once with `bge-small-en-v1.5` and passed to `SketchMemory.ingest`, which updates two stores:

- **Hot memory** (searched at query time; size fixed by the configuration):
  - **Semantic cells.** These are clustering features: embedding sum and count, from which the centroid and radius follow (for unit vectors $E||x-\mu||^2 = 1 - ||\mu||^2$), plus a time span. A chunk joins the most similar cell if cosine ≥ 0.90 (with a recency bonus for the current cell, in the spirit of EM-LLM's event continuity). Otherwise it opens a new cell. When the cell count exceeds `max_cells`, the pair with the smallest **Ward** merge cost $n_a n_b/(n_a+n_b)\cdot||\mu_a-\mu_b||^2$ is merged (merge-and-reduce).
  - **Per-cell Count-Min Sketch** over content terms (width 512, depth 3, conservative update). CMS is linear, so merging two cells just adds their tables.
  - **Extractive digest** per cell: at most 120 words of the most salient sentences. Salience is the mean streaming IDF of a sentence's terms, scaled by how typical its chunk is of the cell. No LLM is called at ingest time.
  - **Exemplars:** up to 8 diverse member embeddings per cell (fp16), kept by dropping the most redundant member. They play the role of EM-LLM's representative tokens when cells are ranked.
  - A global **Count-Min Sketch of document frequencies** (streaming IDF) and a global **Misra-Gries** table of the k = 256 heaviest terms (stream topics).
- **Cold storage** (append-only, never searched; read only on fallback): raw chunk text, its embedding, and the cell→chunk pointer lists. Cold storage grows with the stream, and the report always states which of the two memories a number refers to.

Query time

1. **Cell ranking:** the score is the maximum cosine similarity of the query over each cell's centroid and exemplars. Top-k = 4 cells are taken, plus one temporal neighbour of the best cell (EM-LLM's contiguity buffer). A Count-Min lexical term is available, but dev-split tuning set its weight to 0 for ranking (see Ablations).
2. **Gate.** For the rare query terms (streaming IDF ≥ 2.5), a retrieved cell is escalated to raw evidence if any of these hold:
   - *lossy:* the CMS shows the term in the cell above its noise floor N/w, but the digest lacks it, or no single digest sentence contains the rare terms together;
   - *loose:* the cell radius is above 0.75;
   - *unresolved:* no retrieved cell holds any rare term above the guaranteed bound εN. In that case the top two cells are read raw.

   Questions with no rare terms (global or summary questions) are answered from the sketch.
3. **Raw fetch:** inside an escalated cell, the cell's cold chunks are reranked by cosine similarity plus IDF-weighted term overlap. The best two replace that cell's digest in the context.
4. The LLM answers from: heavy-hitter topics + digests of trusted cells + raw evidence of escalated cells.

Policies compared (RQ4)

- **sketch:** digests only, never fall back.
- **raw:** always fall back on every retrieved cell.
- **gate:** the deterministic gate above.
- **agent:** the LLM sees the sketch and may call `fetch_raw(cell_id)` or `search_memory(query)` tools (at most 3 rounds).

- **hybrid:** the gate's context, plus the same tools.

3. Experimental setup

Data

The test split uses a fixed-seed sample (seed 0) of 30 questions per task:

- 9 English LongBench tasks: NarrativeQA, Qasper, MultiFieldQA-en, HotpotQA, 2WikiMQA, MuSiQue, GovReport, QMSum, PassageRetrieval-en;
- ∞-Bench Retrieve.PassKey and Retrieve.KV;
- 20 questions from ∞-Bench En.QA (books of about 190K tokens).

All tuning used a disjoint **dev split** (seed 1000, 10 questions per task). Test questions were never used to choose a setting. Each document is fed to every system as a chunk stream.

The **long-stream stress test** concatenates every English LongBench QA and summary document (1,115 documents, 8.5M words, about 11M tokens) into one stream in a fixed shuffled order. It then asks 100 probe questions whose source documents are spread evenly over the stream.

LLM and compared systems

All systems share one LLM served by **Ollama** with greedy decoding, the same prompt template, and the same answer-format instructions. Only the context differs between systems. The main runs use `qwen2.5:7b-instruct` (num_ctx 32,768) on a Kaggle T4. Full context is middle-truncated to fit the window, following LongBench's protocol.

- **Full context:** the whole document, truncated to the window.
- **BM25 RAG:** `rank_bm25` top-5 chunks. This replaces Elasticsearch BM25 with the same scoring algorithm.
- **Dense RAG:** FAISS inner product over bge embeddings, top-5 chunks.
- **SARA-lite:** dense top-20 chunks, compressed after retrieval to a 600-word budget by sentence selection. SARA's learned compression vectors are not reproduced.
- **AMBER-lite:** up to 3 rounds of retrieval in which the LLM updates a memory note (AMU), decides whether to retrieve again (AIC) and filters sentences (MCF), using JSON-schema outputs. AMBER's multi-agent prompts and trained filters are not reproduced.
- **EM-LLM:** the *official* repository and evaluation scripts with its Phi-3.5-mini configuration, run on Kaggle 2×T4 on a 20/10-question subset of the same test questions. Hardware-forced changes are listed in the reproduction section. To compare like with like, SketchRAG and the RAG baselines are also run on the same subset with the same base model (`phi3.5` through Ollama).

Metrics

- **Answer quality:** LongBench's official metrics. QA uses token F1 (exact match is also reported), summaries use ROUGE-L, passage retrieval uses its retrieval score, and PassKey/KV use exact containment.
- **Retrieval quality:** *answer recall*, whether a gold answer string appears in the context sent to the LLM. LongBench lacks gold evidence spans, so this is a proxy, labelled as such.
- **Efficiency:** LLM context tokens and LLM calls per question; hot-memory bytes; ingest time per chunk; retrieval latency; fallback rate.

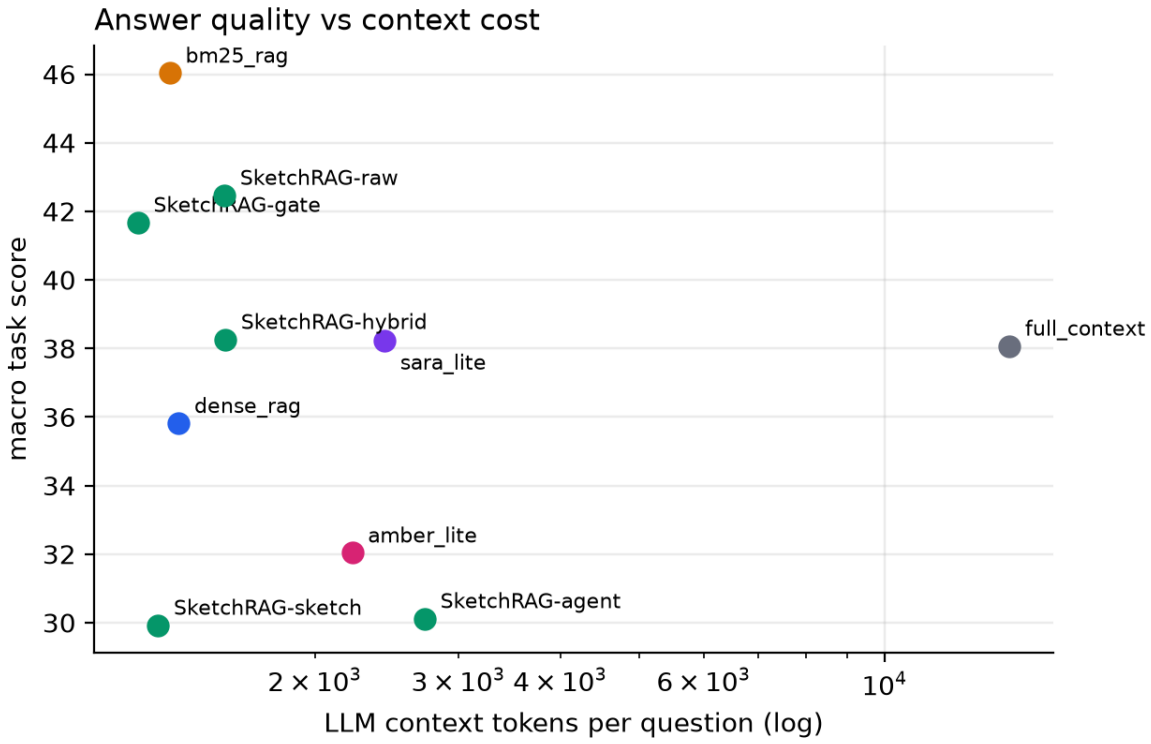
4. Results

4.1 Answer quality (test split, task score ×100)

System	NrtvQA	Qasper	MFQA	HotpotQA	2Wiki	MuSiQue	GovRep	QMSum	PRetr	PassKey	KV	En.QA	Avg
Full context	26.8	51.4	51.7	65.6	58.7	24.2	15.9	19.4	93.3	16.7	10.0	23.0	38.1
BM25 RAG	18.7	45.8	47.5	51.4	38.0	17.6	11.7	20.6	70.0	100.0	96.7	34.3	**46.0**
Dense RAG	14.7	38.8	51.4	47.2	38.3	27.4	11.7	20.0	50.0	100.0	6.7	23.5	35.8
SARA-lite	17.2	53.5	55.0	61.4	51.9	33.5	12.0	18.8	16.7	100.0	16.7	22.0	38.2
AMBER-lite	11.1	42.6	53.3	44.8	43.1	32.9	9.9	18.9	0.0	100.0	10.0	18.0	32.0
SketchRAG · sketch	13.2	24.1	36.5	39.9	28.3	18.6	11.8	17.5	46.7	100.0	6.7	16.0	29.9
SketchRAG · gate	15.0	36.2	44.0	50.2	33.3	21.9	11.7	18.9	53.3	100.0	96.7	19.0	41.7
SketchRAG · raw	20.3	38.6	50.7	52.8	28.5	18.8	13.2	19.4	53.3	100.0	96.7	17.5	42.5
SketchRAG · agent	8.9	20.5	36.0	32.8	20.6	12.3	12.3	18.1	30.0	100.0	60.0	10.0	30.1
SketchRAG · hybrid	14.3	33.1	41.8	47.5	22.6	17.9	12.4	18.7	40.0	100.0	93.3	17.6	38.3

4.2 Efficiency (mean per question)

System	LLM ctx tokens	LLM calls	LLM s	searchable memory (KB)	answer recall	fallback rate
Full context	14232	1.00	61.5	149	0.51	–
BM25 RAG	1327	1.00	8.1	259	0.49	–
Dense RAG	1361	1.00	6.9	447	0.41	–
SARA-lite	2435	1.00	12.9	447	0.45	–
AMBER-lite	2223	2.97	31.8	447	0.33	0.00
SketchRAG · sketch	1284	1.00	6.1	454	0.35	0.00
SketchRAG · gate	1213	1.00	5.8	454	0.46	0.54
SketchRAG · raw	1549	1.00	7.6	454	0.50	1.00
SketchRAG · agent	2727	1.16	9.4	454	0.40	0.06
SketchRAG · hybrid	1553	1.02	6.3	454	0.46	0.55

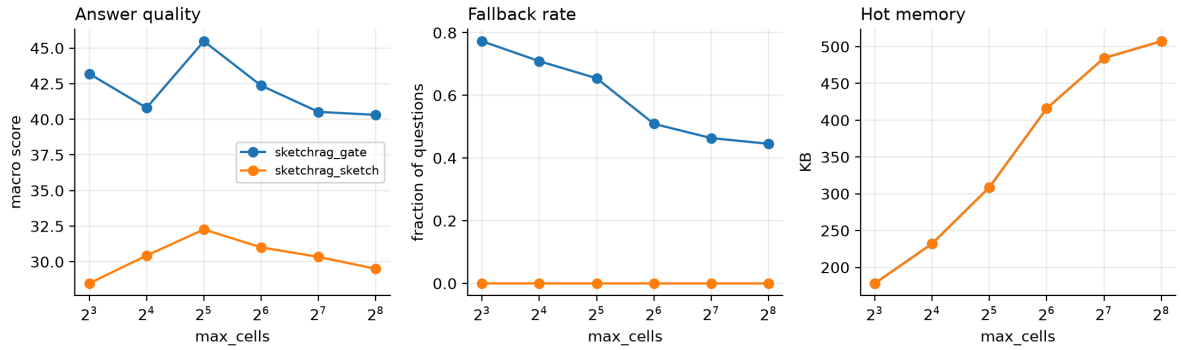


Answer quality versus LLM context tokens per question.

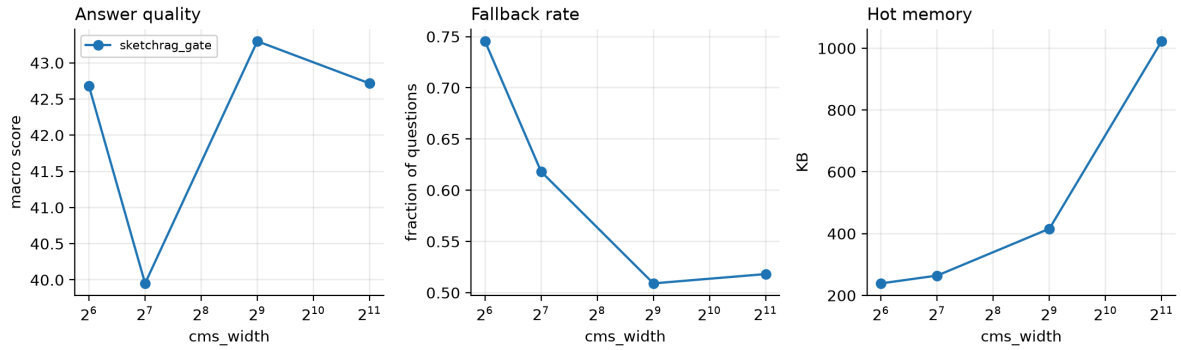
4.3 Ablations of SketchRAG · gate

Variant	Avg score	LLM ctx tokens	fallback rate	n
SketchRAG · gate (frozen config)	42.5	1161	0.51	110
bounded reservoir of 16 pointers per cell	28.9	1165	0.51	110
gate uses empirical noise floor instead of εN bound	42.7	1195	0.61	110
− temporal contiguity	40.7	1011	0.48	110
− co-occurrence check in gate	42.5	1158	0.49	110
raw appended instead of replacing digest	41.5	1645	0.51	110
+ CMS evidence in cell ranking (w=0.5)	41.7	1182	0.54	110

4.4 Memory budget (RQ3, RQ5)



Score, fallback rate and hot memory as the cell budget varies.

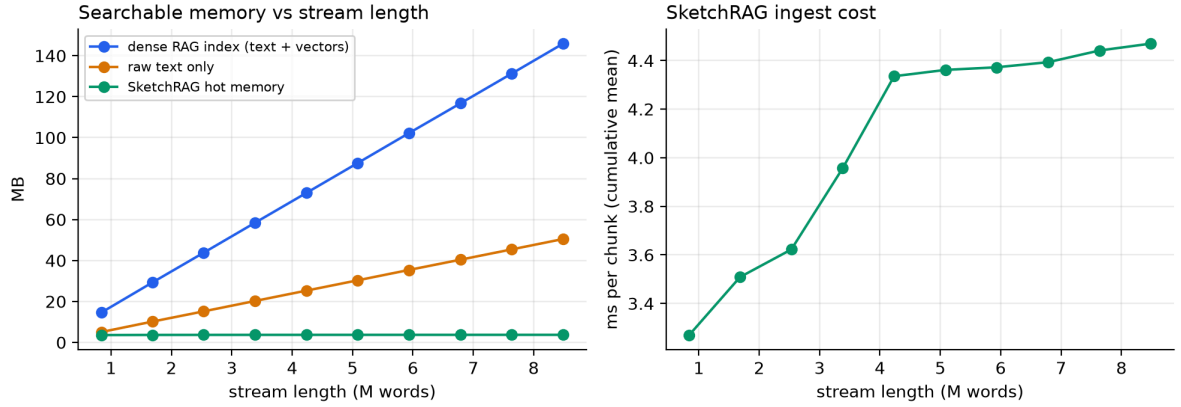


Effect of the Count-Min width (the ε of the gate's bound).

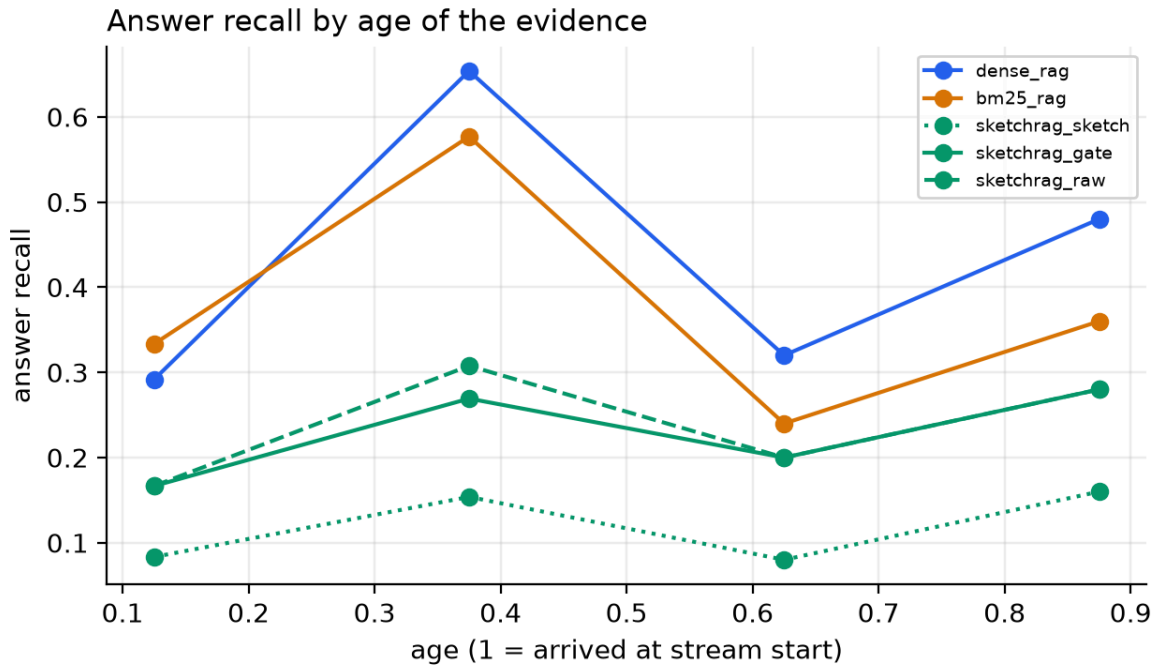
4.5 Long-stream stress test (~11M tokens)

Stream: 65,127 chunks, 8,488,204 words, 100 probe questions. Final dense-RAG index (text + float32 vectors): **146 MB**; raw text alone: 50 MB.

max cells	hot memory (MB)	ingest ms/chunk	answer recall: sketch	gate	raw	dense RAG	BM25 RAG	gate fallback
64	1.01	2.3	0.03	0.29	0.31	0.44	0.38	1.00
256	3.71	4.5	0.12	0.23	0.24	0.44	0.38	1.00



Searchable memory and ingest cost as the stream grows (256 cells).



Answer recall by how long ago the evidence arrived (256 cells).

4.6 EM-LLM reproduction

Category	Tasks run	Ours (official code, subset)	Paper (full LongBench)
SQA	Qasper, MFQA (of 3)	47.1	31.8
MQA	HotpotQA, 2Wiki, MuSiQue	32.7	31.9
Ret	PRetr	15.0	13.0

Same base model (Phi-3.5-mini), same questions:

System	Qasper	MFQA	HotpotQA	2Wiki	MuSiQue	PRetr	Avg
Full context	39.7	41.6	38.3	35.0	11.5	60.0	**37.7**
BM25 RAG	31.4	55.0	42.4	12.1	26.7	40.0	34.6
Dense RAG	37.5	45.0	46.3	29.8	27.7	17.5	34.0
SketchRAG · gate	40.3	47.4	47.3	16.2	20.0	37.5	34.8
SketchRAG · raw	34.1	40.3	56.4	18.6	24.5	43.5	36.2
EM-LLM (official)	43.2	50.9	32.9	27.7	37.5	15.0	34.5

Changes forced by the hardware (2×T4, 29 GB RAM):

- model weights loaded in float16 instead of torch_dtype=auto (bfloat16): T4 has no bf16 support
- min_free_cpu_memory 100 -> 4 (GB): Kaggle VM has ~29 GB RAM
- min_block_size 8 -> 16 and max_cached_block 384 -> 200 (n_mem=3072 kept): the per-layer GPU block cache (max_cached_block x 64 tokens x 32 heads) needed 9.6 GB and ran out of memory next to the weights on a 15 GB T4
- one job with the model split over both T4s (the repo's use_hf_acc path) instead of one job per GPU; weights pre-downloaded to a local folder because that path needs one
- data: fixed-seed subset (20 per LongBench task, 10 per InfiniteBench task) matching the SketchRAG test split
- min_free_cpu_memory 100 -> 8 (GB) with the repo's disk offload enabled (--allow_disk_offload True): Kaggle VM has ~29 GB RAM and run v6 was OOM-killed on NarrativeQA without offload

5. Answers to the research questions

RQ1: Can approximate streaming summaries preserve enough information for accurate RAG? *Partly, and the gate is what makes the difference.*

- **Summary alone.** The digest-only *sketch* policy scores 29.9. It loses most on questions that need an exact string: KV retrieval 6.7, Qasper 24.1.
- **With the gate.** Escalating only the cells whose sketch cannot vouch for the query terms raises the score to 41.7. Always escalating (*raw*) gives 42.5.
- **Global questions.** The summary is sufficient on its own here: GovReport has no rare query terms, the gate never fires, and all policies score alike (11.7–13.2).

So extractive streaming summaries preserve topical content, but exact facts still need the raw text that the sketch points to.

RQ2: How does SketchRAG compare with episodic-memory, adaptive-memory and compression approaches?

- **Against SARA-lite, AMBER-lite and full context.** With the shared 7B model, SketchRAG·gate (41.7) beats SARA-lite (38.2), full context (38.1) and AMBER-lite (32.0). It sends the LLM about half the tokens of SARA-lite and AMBER-lite, with one LLM call per question against AMBER-lite's three.
- **Where it trails.** SARA-lite and full context are clearly better on multi-hop QA (2WikiMQA 51.9 / 58.7 vs 33.3), where evidence is spread over more chunks than the top-4 cells hold.

- **AMBER-lite.** It suffers on PassageRetrieval (0.0) because its memory notes drop the paragraph labels the answer must name. That is a weakness of our re-implementation, not necessarily of AMBER.
- **BM25 RAG.** A plain BM25 baseline is the best system overall (46.0), mainly through exact-match tasks (PassageRetrieval 70.0). That shows how much these benchmarks reward lexical matching.
- **Against EM-LLM.** With Phi-3.5-mini as the shared model, SketchRAG·gate and EM-LLM's official code score within a point of each other on the six tasks EM-LLM completed (Section 4.6). SketchRAG needs one 7B generation per question instead of streaming the whole document through the model. EM-LLM ran in fp16; our Phi-3.5 is Ollama's quantized build, so this is a like-for-like question set, not a like-for-like model.

RQ3: How does reducing memory size affect retrieval accuracy and downstream reasoning?

- **Per document.** Fewer cells mean coarser digests and more fallbacks (Section 4.4, 110 test questions). Going from 256 to 8 cells cuts hot memory from 519 to 183 KB. Sketch-only answer recall drops from 0.37 to 0.28, and the gate compensates by escalating more often (fallback 0.45 → 0.77). The gated score stays between 40.3 and 45.5 at every budget, with no monotone trend.
- **Noise between runs.** Re-running the frozen configuration gives 42.4 and 43.3 on the same 110 questions (42.5 in the main run), so differences of about 2 points in the sweeps are within run-to-run noise.
- **Count-Min width.** A narrower sketch has a looser εN bound, so the gate trusts fewer cells: fallback rises from 0.51 (width 512) to 0.75 (width 64), while the score stays within noise (40.0–43.3). Width 2048 more than doubles hot memory (1,047 vs 426 KB) without lowering the fallback rate.
- **Hot memory at document scale.** A single LongBench document is small enough that 64 cells with exemplars take about as much space as a dense index (454 vs 447 KB). The bounded memory only pays off on long streams.
- **Hot memory at stream scale.** On the 8.5M-word stream the hot memory stays at 1.0 MB (64 cells) or 3.7 MB (256 cells), against 146 MB for dense RAG. Recall falls to 0.29 / 0.23 against 0.44 for dense RAG.
- **Behaviour at scale.** Every probe falls back (fallback rate 1.0), so SketchRAG behaves like a two-level IVF-style router over cold storage. The cell centroid is too blunt a key once thousands of unrelated documents share 64–256 cells.

RQ4: Can an agent decide when the summary is sufficient and when raw evidence is needed? *Yes when the decision is driven by the sketch's error bound; no when it is left to a 7B LLM.*

- **The deterministic gate** escalates 54% of questions.
  - On those, raw evidence lifts the score from 30.0 (digest only) to 51.5.
  - On the 46% it trusts, always reading raw would have added only 0.8 points (31.4 → 32.2).
  - It fires on 100% of KV and 93% of PassKey questions, where digests drop exact keys. It never fires on GovReport.
  - Net effect: it recovers 94% of the always-raw gain while sending 22% fewer context tokens than always-raw (1,213 vs 1,549).
- **The LLM-driven agent** called a tool on only 30 of 350 questions, most of them KV.
  - It stays near the digest-only score (30.1).
  - Hybrid (gate plus tools) adds tokens and scores below the gate alone (38.3), most likely because the extra tool output distracts the 7B model.

The calibrated signal therefore comes from the Count-Min bound, not from the LLM's self-assessment.

RQ5: What is the best trade-off between memory, compression, latency and accuracy?

- **Accuracy versus context cost.** SketchRAG·gate is on the Pareto front (Figure, Section 4.2): the cheapest context of any system (1,213 tokens, 5.8 s of LLM time per question) for the third-best score.

- **Full context** costs 11.7× more tokens and 10.6× more LLM time for a lower score. It also fails PassKey (16.7) because middle truncation removes the needle.
- **Ingest cost.** Ingest takes about 20–26 ms per chunk inside a single document, including the digest and exemplar updates, and 2–5 ms per chunk on the long stream.
- **Recommended setting.** The frozen configuration (64 cells, CMS width 512, gate with two raw chunks per escalated cell) is the best balance we found. More cells cost hot memory without improving test recall at stream scale. On single documents, 8–32 cells cost no accuracy and use 40–60% less hot memory, but they send the agent to cold storage more often.
- **What must not be cut.** The ablations (Section 4.3) show that the raw pointers matter most. Capping each cell at a 16-pointer reservoir drops the gated score from 42.5 to 28.9, the digest-only level, because the raw chunk holding the answer is often no longer reachable. Removing temporal contiguity costs 1.8 points. The other variants are within noise.

6. Limitations

- **The hot memory is bounded; cold storage is not.** The fallback needs raw text on disk. The *sketch* policy is the only truly bounded configuration end to end, and it is reported separately.
- **Re-implementations, not reproductions.** AMBER and SARA are lightweight re-implementations on a shared LLM. Only EM-LLM is run from its official code, and only at reduced scale on T4 GPUs with the changes listed. Its ∞-Bench runs (KV, PassKey; ~128K-token inputs) and NarrativeQA (killed after its first question, twice) exceeded the 29 GB host-RAM limit even with disk offload, so EM-LLM is reported on 6 of the 9 planned tasks.
- **Answer recall is a proxy.** It is a string-containment proxy for retrieval quality, not Recall@K against annotated evidence.
- **Single open model.** One 7B model (plus Phi-3.5 for the matched comparison) and 30 questions per task. Small differences between systems (<3 points) are within noise.

7. Reproducibility

Code: `sketchrag/` (unit tests: `pytest`). Every table and figure in this report is regenerated from `results/` by `report/build_report.py`; the dev-split tuning log is in `results/tuning_dev.md`.